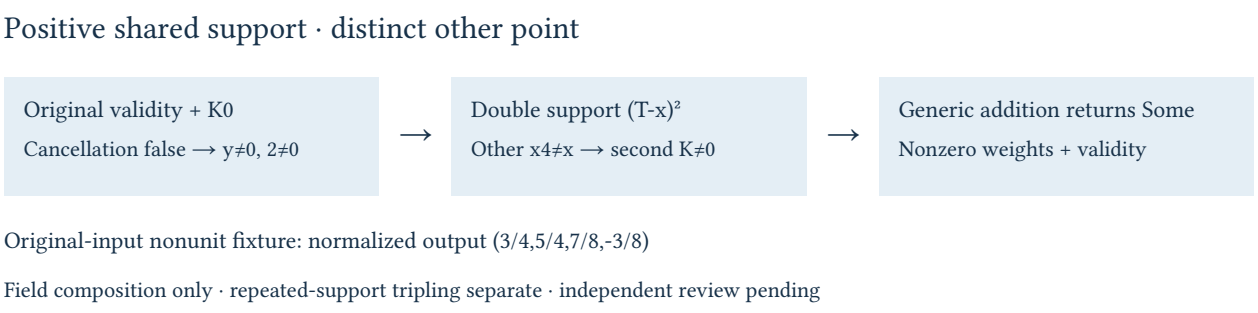
Positive shared-support distinct-other composition



Derived a field-level composition for the positive shared-support, distinct-other-point branch of add12_weighted_complete. The main module, full original-input controls and separate axiom readback all exited0. Seven named declarations use only propext, Classical.choice and Quot.sound. Independent source-fidelity review remains pending.

The final theorem source_branch_preserves assumes only the original degree1 and degree2 normalized validity predicates on the monic quintic X5+a3X3+a2X2+a1X+a0, four nonzero representation weights, outer rawK=0, cancellation guard false and the source distinct-other-support test. It derives positiveCoincidence, characteristic not2, nonzero original y, successful generic Some return, nonzero final weights and normalized degree2 output validity. Positive ordinate equality, doubling validity and second genericK nonzero are derived, not added as output assumptions.

At rawK=0, the reviewed extraction theorem derives the two ordinate alternatives from original validity. Cancellation failure removes the negative alternative. Write L=a.y*rawW3 and R=rawY3*a.w. The remaining L=R and L!=-R give L!=-L, which rules out both 2=0 and a.y=0. Thus this path supplies exactly the nonzero prerequisites of the reviewed doubling invariant without adding a characteristic hypothesis.

The source comparison a.x*b.z² != extracted.x implies distinct normalized support coordinates after multiplying by the shared nonzero normalization denominator. The reviewed doubling formula has normalized support $T^2-2xT+x^2=(T-x)^2$. At the extracted other coordinate x4 its value is $(x4-x)^2$, nonzero by that source comparison. The reviewed scaling identity then proves genericK(extracted,doubled) nonzero. Generic addition therefore returns Some; its validity theorem composes with extracted-point and doubled-divisor validity.

The bridge from original degree1 divisibility to the doubling point equation is proved by explicit division remainder: $f-y^2=(T-x)*Q+(f(x)-y^2)$. A degree1 polynomial dividing the constant remainder forces it to vanish. This avoids assuming the doubling point equation separately. No class-sum or homomorphism conclusion is used.

The full rational nonunit fixture uses a=(0,64,2,2),b=(-9,0,0,486,3,2),f=X5-X+1. It normalizes to point(0,1), supportT(T-1), ordinate1. Both original validity witnesses are explicitly constructed, and the full source theorem is invoked. The literal final normalized output is(q,r,s,t)=(3/4,5/4,7/8,-3/8). Boundary controls check zero-ordinate cancellation, equal-other-support positive coincidence excluded from this branch, and an invalid original point by its constant residual. These are fixed polynomial fixtures, not smooth-curve claims.

The repeated-support case selects the separate source tripling path and remains outside this theorem. The result concerns retained field expressions, including doubling output extraction and the nested generic call. It does not establish literal UInt64 correspondence, counter or compiler behavior, divisor-class addition, or complete mixed dispatch/walk closure. No scientific trial or timing measurement was run.

Attempts1-3 are retained: the first used an unavailable polynomial eval API and an unnecessary trailing tactic; the second exposed overbroad simp expansion of constant polynomial degree/coefficients; the third normalized the positive guard before using its inequality. The final proof uses explicit constant remainders, narrowly scoped polynomial simplification and source-order guard rewriting. These were proof-construction failures; fixture/claim scope did not change.

Source links and hashes bind mixed dispatch828-878, doubling477-510 and generic232-292, plus the reviewed extraction, doubling and generic proof dependencies. Each Lean compile had a120second bound and used existing pinned dependency oleans. No independent binary/kernel rechecker is claimed. Original packets remain unchanged; this additive packet is ready for independent review.